Muhammad Raihan Thaffan Hidayat

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Profile

Final Year Telecommunications Engineering Student at Telkom University (8th Semester). Has practical experience in developing IoT solutions using ESP32 and Arduino, as well as front-end web development with React.js. Currently active as a Teaching Assistant at the MBC Laboratory, responsible for module development and technical training. Highly committed to continuous learning and applying technical expertise to innovative projects.

Work Experience

Project Management Officer, Internship **Jun 2025 - Sept 2025**
PT LEN Industri (Persero), Bandung

- Developed and managed project dashboards using Google Looker Studio for dynamic, real-time project monitoring with a focus on budget oversight.
- Utilized AppSheet and Looker to optimize project management processes during the internship period.
- Prepared various project documents required for management reporting.

Project

ESP32 Based Smart Assistive Device for the Visually Impaired **Mar 2025 - Jun 2025**

- Designed a navigation aid for the visually impaired using an ESP32 microcontroller to enhance user safety and mobility.
- Integrated ultrasonic sensors for obstacle detection, a tilt sensor for balance monitoring, and a buzzer alarm system.
- Developed device logic using C++ and integrated it with the Blynk.io platform for monitoring functionality.

CryptoMania Website Development **Mar 2025 - Jun 2025**

- Built a cryptocurrency information platform providing real-time data on coin names, market values, descriptions, and price history.
- Led the front-end development using React.js, JavaScript, and CSS.
- Designed and implemented a responsive user interface to present data for the top 50 coins informatively.

Organizational Experience

Public Relations Division, PERMISI **2022 - 2023**

- Actively participated as a team member in the Education Roadshow project targeting various high schools in the Bekasi area.
- Carried out intensive coordination and direct socialization of the program to schools to build and maintain strategic cooperative relationships.
- Managed internal team communication to ensure that activity schedules and presentation materials were delivered effectively and on time.
- Achievements: Successfully reached and increased the enthusiasm of high school students in Bekasi in learning about Telkom University's education programs through measurable outreach activities

Public Relations Division, TELERIUM **2023 - 2024**

- Acting as a member who serves as the main communication bridge through intensive coordination between laboratories within the Faculty of Electrical Engineering.
- Ensuring the smooth flow of information between the organizers and laboratory assistants in order to minimize technical obstacles during the event.

Practicum Division, MBC LABORATORY **2025 - Present**

- Responsible for teaching practical material directly to students in the laboratory to ensure in-depth technical understanding.
- Compiling and developing practical modules as the main technical guide in the student learning process.
- Implementing the Training of Trainers (ToT) program to provide teaching material training to fellow assistants to ensure the quality of instruction before the practical period begins.
- Installing and configuring the necessary software to support the smooth running of practical activities.
- Managing the physical maintenance of the laboratory, including workbench layout and cable management, to ensure that the practical area remains safe and conducive.

Education

Telkom University **2022 - Present**
S1 Teknik Telekomunikasi

Soft Skill

- Time Management
- Discipline
- Adaptability
- Teamwork

Hard Skill

- IoT Development: Arduino, ESP32
- Sensor Integration: Ultrasonic, Buzzer, Tilt Sensor
- Programming Languages (basic): C/C++, Python, HTML, CSS, JavaScript, MySQL
- Networking: Basic network configuration using Cisco Packet Tracer
- Microsoft Excel (basic)
- Google Looker Studio

Licenses & Certifications

Learn the Basics of AI, Dicoding Indonesia — Nov 2024 - Nov 2027

Getting Started with Python Programming, Dicoding Indonesia — Nov 2024 - Nov 2027

Learn the Basics of Data Science, Dicoding Indonesia — Oct 2024 - Oct 2027

Learn the Basics of Structured Query Language (SQL), Dicoding Indonesia — Oct 2024 - Oct 2027